

Advanced (Carolina Reaper)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
- (B) $ax + bx^2 + c$
- (C) $ax^2 - bx + c$
- (D) $ax^3 + bx^2 + c$
- (E) $ax^2 + b$

Q2. Which of the following is a quadratic function in standard form?

- (A) $x^2 + 3x - 2$
- (B) $x^3 - 2x^2 + x$
- (C) $x^2 - 4$
- (D) $2x^4 - x^2 + 1$
- (E) $x - 2$

Q3. What is the coefficient of the x^2 term in the quadratic $3x^2 + 2x - 1$?

- (A) 1
- (B) 2
- (C) 3
- (D) -1
- (E) -2

Q4. Which of the following quadratic functions has a leading coefficient of -2?

- (A) $-2x^2 + x - 1$
- (B) $2x^2 + x - 1$
- (C) $-x^2 + 2x + 1$
- (D) $x^2 - 2x - 1$
- (E) $-3x^2 + x + 1$

Q5. What is the value of b in the quadratic $x^2 + bx - 3$ if the quadratic has a root at $x = 2$?

- (A) -1
- (B) 1
- (C) -5
- (D) 5
- (E) 3

Q6. Which of the following is NOT a key feature of a quadratic function?

- (A) Vertex
- (B) Axis of symmetry
- (C) Roots
- (D) End behavior
- (E) Period

Q7. What is the axis of symmetry of the quadratic $x^2 + 4x + 4$?

- (A) $x = -1$
- (B) $x = -2$
- (C) $x = 0$
- (D) $x = 2$
- (E) $x = -4$

Q8. Which of the following quadratics has an axis of symmetry at $x = -3$?

- (A) $x^2 + 6x + 9$
- (B) $x^2 - 6x + 9$
- (C) $x^2 + 2x - 3$
- (D) $x^2 - 2x - 3$
- (E) $x^2 + 4x + 4$

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic $x^2 - 4x - 3$. Be sure to include the axis of symmetry, vertex, and roots.

Q10. Find the equation of a quadratic with roots $x = 2$ and $x = -1$, and leading coefficient 2. Show all work and justify your answer.

- Use a graphing calculator to check student answers.
- Emphasize the importance of showing work and justifying answers.
- Remind students to use the standard form of a quadratic when solving problems.